

NOATAK, AK, USA

WMO: 701335

Lat: 67.566N

Lon: 162.975W

Elev: 27

StdP: 101.00

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 26648

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-37.4	-34.2	0.2	-30.4	-33.7	0.2	-29.3	12.9	-6.8	11.7	-7.0	0.7	10	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	23.9	14.7	22.1	13.9	19.3	12.8	15.6	22.3	14.4	20.3	13.4	18.7	3.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.1	17.8	11.4	8.4	16.4	10.3	7.8	15.1	43.5	22.2	40.3	19.7	37.3	18.6	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.3	8.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-34.1	17.0	2.0	0.9	-35.6	17.7	-36.8	18.2	-37.9	18.7	-39.3	19.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	-19.4	-16.0	-14.9	-7.9	2.7	11.3	12.2	9.8	5.3	-3.9	-13.8	-16.4
	DBStd	13.37	10.73	9.53	7.09	6.66	5.82	4.31	3.48	3.13	3.98	6.38	7.38	8.64
	HDD10.0	5382	911	728	772	538	232	35	17	41	145	433	714	817
	HDD18.3	8220	1169	962	1030	788	484	213	190	266	390	691	964	1075
	CDD10.0	209	0	0	0	0	7	75	87	34	5	1	0	0
	CDD18.3	5	0	0	0	0	0	3	2	1	0	0	0	0
	CDH23.3	87	0	0	0	0	2	35	39	7	0	4	0	0
	CDH26.7	12	0	0	0	0	0	4	5	0	0	2	0	0
Wind	WSAvg	3.3	3.1	3.9	3.5	3.7	3.3	3.1	2.9	3.1	3.3	3.5	3.2	3.4
Precipitation	PrecAvg	452	32	35	26	21	17	20	49	72	50	60	39	31
	PrecMax	630	68	111	126	55	46	37	98	155	111	116	116	78
	PrecMin	324	7	5	6	3	1	9	12	14	13	24	6	10
	PrecStd	79	16	26	24	15	12	8	20	33	21	24	25	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	2.4	2.0	7.8	22.2	27.1	27.3	23.9	17.4	12.2	2.5	2.6
		MCWB	0.4	0.5	0.4	3.5	11.9	15.3	16.5	16.4	10.8	5.8	0.7	0.3
	2%	DB	-0.3	0.3	-0.2	4.9	17.6	23.9	24.0	19.2	14.8	7.1	1.1	1.0
		MCWB	-1.3	-0.6	-1.2	1.7	10.1	14.1	15.2	13.5	9.2	4.3	-0.2	-0.6
	5%	DB	-2.2	-1.2	-2.2	2.8	14.8	22.2	22.2	17.4	12.8	4.4	-1.1	-2.0
		MCWB	-3.2	-2.0	-3.7	0.4	8.5	13.3	14.5	12.2	8.8	2.2	-1.7	-3.0
	10%	DB	-3.9	-2.7	-3.8	1.7	12.1	19.9	19.7	15.9	11.3	3.0	-2.8	-4.4
		MCWB	-5.1	-3.7	-5.1	-0.2	7.1	12.2	13.9	11.4	8.3	1.0	-3.9	-5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	0.6	0.6	3.7	12.4	16.4	17.1	17.2	11.7	6.8	1.1	1.0
		MCDB	1.7	1.8	1.5	7.6	20.7	25.1	24.7	23.7	14.6	10.5	2.2	2.3
	2%	WB	-1.1	-0.3	-1.4	2.0	10.4	14.8	15.9	14.4	10.6	4.5	-0.2	-0.6
		MCDB	-0.1	0.6	-0.3	4.5	17.2	22.4	22.7	18.1	13.2	7.0	0.6	0.4
	5%	WB	-2.5	-2.0	-3.5	0.9	8.8	13.7	15.0	13.1	9.6	3.0	-1.8	-2.7
		MCDB	-1.8	-1.4	-2.3	2.8	14.5	20.8	21.2	16.7	12.6	4.3	-1.1	-1.6
	10%	WB	-4.4	-3.5	-4.9	-0.1	7.1	12.5	13.9	11.8	8.5	1.6	-3.9	-5.1
		MCDB	-3.2	-2.5	-3.5	1.5	11.5	19.2	19.0	15.0	11.0	3.1	-2.9	-4.0
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.2	9.2	9.2	9.1	10.6	9.7	8.4	8.0	6.5	6.5	7.7
		MCDBR	7.9	7.3	8.8	8.6	14.3	13.6	15.1	12.0	10.4	7.2	5.3	7.5
		MCWBR	7.0	6.9	7.8	6.6	8.3	6.6	7.8	7.3	6.6	5.1	4.9	6.6
	5% WB	MCDBR	7.5	7.4	8.0	8.1	13.6	12.5	13.1	10.4	8.8	7.1	5.4	7.0
		MCWBR	6.9	7.1	7.2	6.3	8.1	6.6	7.1	6.6	5.7	5.2	5.1	6.3
Clear-Sky Solar Irradiance	taub	0.174	0.241	0.261	0.325	0.369	0.341	0.348	0.357	0.309	0.245	0.168	N/A	
	taud	1.903	2.088	2.226	2.096	2.165	2.443	2.450	2.424	2.502	2.317	1.942	N/A	
	Ebn at Noon	154	576	795	814	816	864	844	788	751	593	150	0	
	Edh at Noon	11	49	76	116	122	95	92	85	61	38	9	0	
All-Sky Solar Radiation	RadAvg	0.05	0.58	2.34	4.55	5.51	5.99	4.77	3.50	2.15	0.77	0.12	0.00	
	RadStd	0.01	0.07	0.20	0.26	0.63	0.55	0.51	0.44	0.25	0.09	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page